

Module 003 Bridge Twins

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Abstract—This module forms a critical component of the CEREBRUM framework, a community-structured graph attention engine for Knowledge Graph reasoning. It focuses on Advanced Graph Attention Analysis.

Index Terms—Knowledge Graph, Graph Attention, DSCF, CSA, Hierarchical Reasoning.

1 EXPERIENCE-DEPENDENT STRUCTURAL PLASTICITY IN KNOWLEDGE GRAPHS: THE BRIDGE TWIN ENGINE

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1.0.1 Abstract

We introduce the **Bridge Twin Engine**, a mechanism for autonomous topological evolution in Knowledge Graphs (KGs) based on reasoning experience. While traditional KGs maintain a static structure, the Bridge Twin Engine implements a graph-theoretic analog to biological Long-Term Potentiation (LTP) [?], [?]. By materializing "Twin Nodes" to act as structural relays across frequently traversed community boundaries, the system short-circuits latent paths and optimizes its own geometry for future inference. We formalize the potentiation rules based on usage frequency and semantic salience, and provide a pruning mechanism (LTD) to maintain topological sanity. The v1.2.0 release incorporates an atomic re-mapping protocol to synchronize relays with background community re-partitioning, effectively solving the "Zombie Bridge" problem in dynamic environments.

1.0.2 1. Introduction

The efficiency of multi-hop reasoning in graphs is fundamentally constrained by the diameter and sparsity of the network. In large-scale KGs, semantically related concepts often reside in disparate community partitions, necessitating high-latency "Bridge" traversals. We propose that a graph should not be a static artifact, but a plastic entity that reshapes its topology to favor successful reasoning chains.

1.0.3 2. Methodology

1.0.3.1 2.1 The Potentiation Rule: A structural relay node v'_{twin} is materialized for node v in destination community C_{dest} if:

- 1) **Frequency:** Traversal ($u \rightarrow v$) occurs $\geq n_{min}$ times.
- 2) **Alignment:** $\cos(\vec{e}_v, \vec{c}_{dest}) \geq \sigma$, where $\vec{c}_{dest}$ is the community centroid.

1.0.3.2 2.2 Materialization and Reflection: Upon potentiation, the engine performs a "Reflection" operation:

- Assign v'_{twin} to C_{dest} .
- Establish a BRIDGE_TWIN binding edge (v, v'_{twin}).
- Mirror all of v 's edges that terminate in C_{dest} onto v'_{twin} .

1.0.4 3. Structural Maintenance: The LTD Analog

To prevent the "Informational Sprawl" of materialized nodes, we implement a temporal decay function:

$$c_t = c_0 \cdot \lambda^{\Delta t} \quad (1)$$

where λ is the decay constant and Δt is the time since last usage. Edges falling below a confidence threshold c_{min} are pruned during the **REM Cycle** (SPEC_007).

1.0.5 4. Conclusion

The Bridge Twin Engine provides a biologically-inspired framework for self-optimizing Knowledge Graphs. By transforming reasoning history into physical graph structure, it enables sub-millisecond multi-hop reasoning on increasingly complex network topologies.

— References

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